

Program: Diploma in Printing Technology	
Course Code: 6109A	Course Title: Specialized Printing Applications Lab
Semester: 6	Credits :1.5
Course Category: Program Elective	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To develop tangible knowledge & experience with various specialized applications of printing.
- To provide functional experience with security printing applications.
- To be familiarize with textile printing with different printing techniques.
- To develop precise skill to perform customization in variable data printing

Course Pre-requisites:

Topic	Course code	Course Name	Semester
Sheet-fed offset printing		Basics of Printing Processes	II
Digital Printing		Digital Imaging Techniques	IV
Screen Printing		Basic Screen Printing Lab	III
Security Ink and paper		Materials for Printing and Packaging.	III

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	To perform security printing.	11	Applying
CO2	To work with digital printing for customized variable data printing.	9	Applying

CO3	To perform the creation and printing of machine readable code.	10	Applying
CO4	To perform textile printing with different printing techniques.	11	Applying
	Lab Exam	4	

CO-PO Mapping:

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped

2-Moderately mapped

1-Weakly mapped

Course Outline:

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	To perform security printing.		
M1.01	Design and Print a security document with security ink.	3	Applying
M1.02	Design and Print a security document on Water mark or Copy evident paper.	3	Applying
M1.03	Design and Print a security document with Micro text, Clear text and Thermo text.	5	Applying
CO2	To work with digital printing for customized variable data printing.		
M1.01	Familiarize with the features and working of inkjet printer.	3	Applying

M1.02	Design and generating of customized design for variable data printing.	6	Applying
	Lab exam I	2	
CO3	To perform the creation and printing of machine readable code.		
M1.01	Generate and print barcode with digital and offset printing.	5	Applying
M1.02	Generate and print QR code with digital and offset printing.	5	
CO4	To perform textile printing with different printing techniques.		
M1.01	Textile printing with Hand block printing, stencil or screen printing, spray printing.	6	Applying
M1.02	Poster design and digital printing on Textile material.	5	Applying
	Lab exam II	2	

Text / Reference:

T/R	Book Title/Author
T1	Helmut Kipphan, Hand book of print media , Springer Science & Business.
R1	Reid Anderson, Exploring Digital Prepress: 1st edition , Delmar Cengage Learning.
R2	Harald Jhonson, Mastering Digital Printing, Second Edition.
R3	Adams J.M, Printing Technology 5th edition , Delmar Publishers, NewYork

OnlineResources:

Sl No	Website Link
1	https://www.securityprinting.co.uk/security-inks.php
2	https://www.mpofcinci.com/blog/barcode-vs-qr-code
3	https://en.wikipedia.org/wiki/Variable_data_printing
4	https://en.wikipedia.org/wiki/Textile_printing